Examiner only

6. Following the Chernobyl nuclear reactor accident in 1986, RIMNET (Radiation Incident Monitoring Network) was installed across the UK to monitor the consequences for the UK of nuclear incidents abroad.
GM tubes are used to measure background levels of radiation.
The permanent monitoring stations in Wales are located in the places shown on the map below.

(a) Suggest why there is **more than one** monitoring station located in Wales. [1]

....

(b) The background count at one of the monitoring stations is 24 counts per minute.
Explain how, using a stopwatch and GM tube connected to a counter, an accurate value for the background count is determined. [2]

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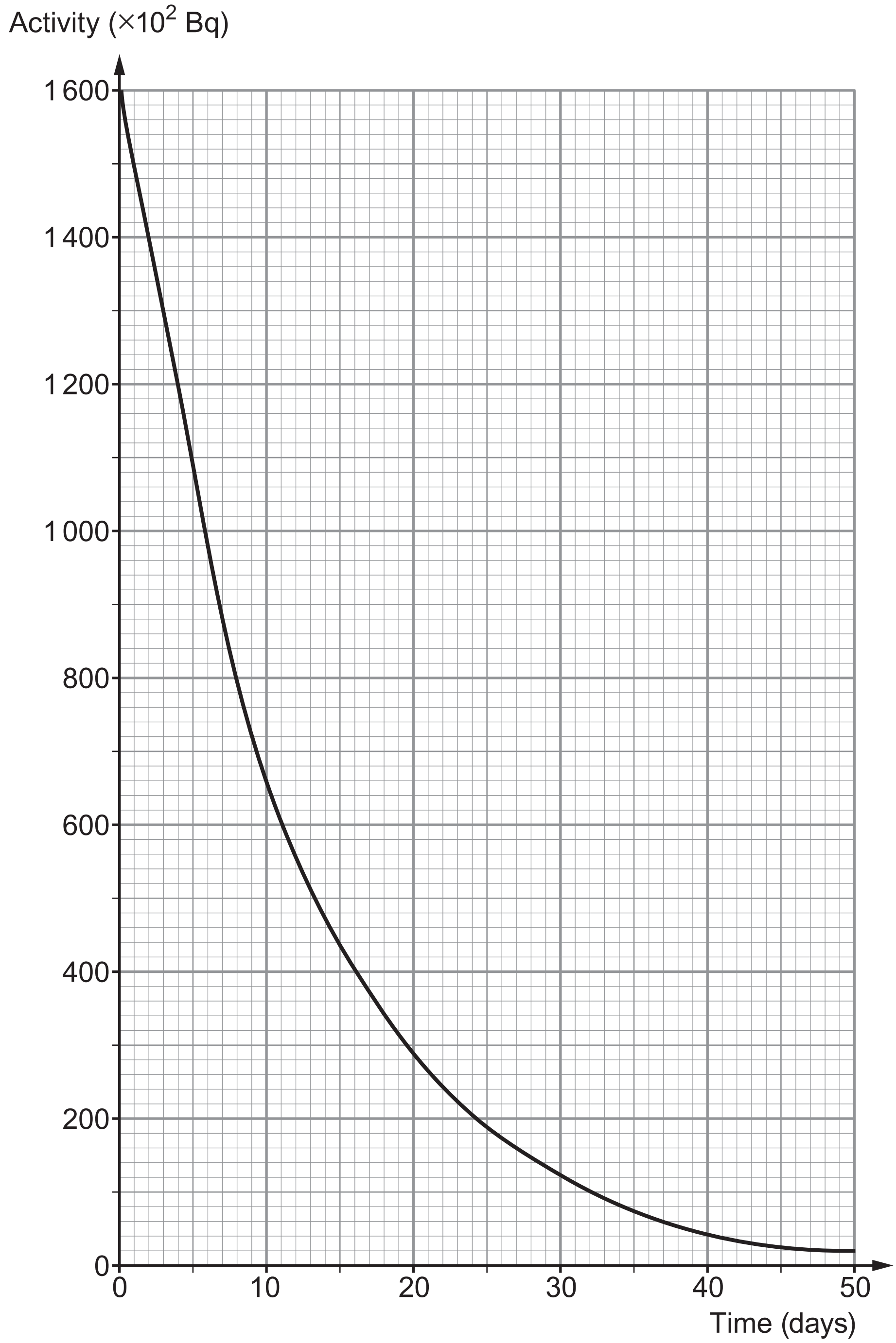
(c) Two radioisotopes released from the Chernobyl accident were iodine-131 (I-131) and caesium-137 (Cs-137).
The decay curve for a sample of I-131 is shown on the graph below.



(i) State the meaning of half-life. [2]

....

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(ii) Use information from the graph to determine the half-life of I-131. [1]

Half life = ... days

(d) A sample of Cs-137, with an activity of 1.5 kBq, was obtained from near Chernobyl. Calculate the time taken for this sample to reach a safe limit of $\frac{1}{512}$ of its original activity. The half-life of Cs-137 is 30.2 years. [3]
Space for calculations.

Time taken = ... years

(e) Explain why in 2021, 35 years after the nuclear accident, scientists were concerned about the Cs-137 contamination in the Chernobyl area but not I-131. [2]

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END OF PAPER

